Build • Learn • Grow

RozgarLink

Skills-to-Opportunity Matching Engine

Product Requirements Document v1.0 | A FutoraOne product by Futora Group
Domain 7 - Government Systems & GovTech | Problem G-01



Eligibility-first | Explainable ranking | Skill pathways | Live updates

THE ENTIRE WORKING OF ROZGARLINK, END TO END

RozgarLink connects **candidate profiles** with **manually entered opportunities**. Its one non-negotiable design rule: **eligibility is decided by deterministic rules stated by the opportunity; AI only helps understand skills and influence ranking.** Every result is explained in plain language, and any change to a candidate updates the screen instantly.

#	Stage	What happens	Who decides
1	Opportunity entry	An officer/admin types in an opportunity: title, work involved, location, **required skills**, **mandatory eligibility criteria** (qualification, residence, age, certification, experience) and **preference attributes** (pay, shift, work type). Criteria are saved as structured rules, never as free text.	Human (officer)
2	Candidate profile	Candidate enters skills (free text allowed), qualifications, location and preferences (commute, pay, work type). AI extracts skills from free text; the candidate confirms them.	Candidate (+AI suggests)
3	Skill normalisation	'MS Excel', 'spreadsheets', 'advanced excel' map to one canonical skill via alias dictionary first, then embedding/LLM similarity with a confidence score. Output is used for ranking only.	AI + dictionary (ranking only)
4	Eligibility gate	A pure rule engine checks every mandatory criterion: PASS, FAIL or UNKNOWN. Eligible only if **all** pass. Skill similarity never enters this step. Failed rules carry the exact condition text.	Rules only (no AI)
5	Ranking	Only eligible opportunities are scored: skill fit 55%, preference fit 30%, proficiency/experience 15%. Semantic matches earn partial, capped credit.	Formula (AI-informed inputs)
6	Explanation	Each eligible result shows why it ranks where it does (score breakdown in words). Each ineligible result names the specific failed condition and the candidate's current value.	Templates from rule/score data
7	Skill gap + pathway	For opportunities the candidate marks 'interested', missing required skills are listed and mapped to courses from the team-supplied catalogue (no live provider access), ordered by coverage, duration and prerequisites.	Catalogue lookup
8	Live recompute	Editing qualification, location, skills or preferences fires an event; the engine re-evaluates immediately; the UI updates in place and highlights what changed ('Now eligible', 'No longer eligible', rank moved).	Engine + push (SSE)

What the candidate sees after matching

Eligible matches (ranked)	Not eligible (with reason)	Need more information
Ordered list with score, 'Why this rank' sentences, skill-gap chip and learning pathway button.	Greyed list, never mixed into ranks. Each card states the failed rule, e.g. 'Requires residence in Maharashtra. Your location: Gujarat.'	Opportunities where a criterion cannot be checked yet because profile data is missing. Prompt: 'Add your qualification to check this.'

Why this matters

Treating a requirement as a preference recommends jobs a candidate cannot formally take. Treating a preference as a requirement hides options they could pursue. RozgarLink separates the two in the data model, the engine, the UI and the tests. Output is decision support; the opportunity's stated criteria always govern.

02 Document Control, Vision and Goals

Item	Detail
Product	RozgarLink - Skills-to-Opportunity Matching Engine (Problem G-01)
Document	PRD v1.0 (20 pages), build-ready for MVP and demo
Owner	Futora Group - Product & Engineering
Audience	Product, engineering, design, QA, government/programme stakeholders, evaluators
Status	Draft for review. Assumptions flagged in Section 20

Contents

Pages	Sections
1-4	Working overview, vision, problem and users, scope and success metrics
5-9	Eligibility vs preference principle, functional requirements, data model, eligibility engine, skill normalisation and AI guardrails
10-14	Ranking, explanations, skill gaps and pathways, real-time updates, UI/UX
15-18	Architecture, API, non-functional requirements, test plan
19-20	Seed data and demo scenarios, roadmap, risks, open questions

Vision

Every job seeker and trainee should see opportunities they can actually take, ranked by how well they fit, with clear reasons, and a concrete path to close gaps for the ones they want.

Product goals

- **G1 Correctness first:** zero ineligible opportunities ever shown as eligible matches.
- **G2 Vocabulary-proof:** same skill in different words resolves to one canonical skill.
- **G3 Explainable:** every result carries a plain-language reason; no black-box scores.
- **G4 Actionable:** skill gaps lead to learning pathways from a supplied catalogue.
- **G5 Live:** profile changes reflect in eligibility and ranking within one second, visibly.
- **G6 Decision support, not decision making:** opportunity-stated criteria are final; humans stay accountable.

Guiding principles

Principle	Meaning in the product
Rules decide, AI assists	AI never touches the eligibility result. It can suggest skills and similarity that adjust rank.
Unknown is not eligible	Missing data yields 'Need more information', never a silent pass or fail.
Show your work	Each decision stores the rule, inputs and outcome so it can be explained and audited.
Plain language	Explanations are readable by someone with no technical or policy background; English first, Hindi/Marathi-ready strings.
Manual-first data	Opportunities are entered by humans in MVP; no scraping, no assumption of external feeds.

03 Problem Statement, Users and Personas

Problem

- Listings describe one skill in many wordings, so keyword search hides qualified candidates.
- Eligibility conditions (must meet) and preferences (affect ranking) are commonly mixed up.
- Candidates do not know *why* an opportunity is missing or what to learn to qualify.
- Profile changes (a new certificate, a move to another district) do not instantly change what is available.

User roles

Role	Description	Key needs
Candidate / trainee	Job seeker or trainee building a profile	Find takeable roles, understand reasons, see gaps, learn
Opportunity officer	Staff who enter and maintain opportunities	Fast structured entry, validation, preview of who would match
Catalogue manager	Team member who maintains skills and learning catalogue	Add skills, aliases, courses, prerequisites
Programme admin	Oversees programme and audit	Dashboards, audit logs, configuration, role control
Evaluator (read-only)	Reviews decision support quality	Trace any result to its rule and inputs

Personas

Persona	Situation	What RozgarLink does
Asha, 22, Pune	ITI Electrician, lists 'wiring, panel fitting, MCB install'	Maps to canonical 'Electrical wiring'; sees 3 eligible roles; one Nashik role marked not eligible (residence rule) with exact reason
Rahul, 27, Nagpur	12th pass, 'tally, gst filing, excel'	Aliases resolve to Accounting software and Excel; diploma-required role shows 'Requires Diploma. You have: 12th pass'; learning pathway suggested
Meena, 34, Kolhapur	Returning to work, tailoring skills, prefers short commute	Preference-only commute filter reorders results but never hides eligible ones
Officer Kiran	Enters 40 apprenticeships a week	Uses a rule builder; sees live count of 'candidates who would be eligible'

Out-of-scope users for MVP

Employers self-serve posting, recruiters bulk-importing candidates, and minors under programme-defined minimum age are not supported in MVP.

04 Scope, Assumptions and Success Metrics

In scope (MVP)	Out of scope (later)
Candidate profile CRUD with skills, qualifications, location, preferences	Application submission and tracking
Manual opportunity entry with structured eligibility rules	Automated scraping or feeds
Skill taxonomy, aliases, AI extraction and similarity	Live course-provider APIs or enrolment
Eligibility engine, ranking, explanations	Resume file parsing (PDF/DOC) beyond text paste
Skill gap analysis and catalogue-based pathways	Predictive hiring outcomes or auto-decisions
Live recompute and visible change highlighting	Native mobile apps (responsive web first)
Role-based access, audit log, English UI with i18n hooks	Full multilingual content

Assumptions

- Opportunities and the learning catalogue are supplied and maintained by the team; volumes up to 10,000 opportunities and 100,000 candidates at MVP.
- A qualification ladder (e.g. 8th, 10th, 12th, ITI, Diploma, Graduate, Postgraduate) is configured by the programme.
- Location is captured as state, district and optional pincode; distance uses district centroids (no live maps needed).
- AI service is an external LLM/embedding API; the system must work, with reduced skill matching, if it is unavailable.

Success metrics

Metric	Target	How measured
Ineligible shown as eligible	0 (hard gate)	Automated rule test suite + production audit sample
Alias recognition accuracy	95% or higher on labelled test set of 300 phrases	Offline evaluation each release
Explanation coverage	100% of results have a reason	Contract test on API response
Recompute latency (profile change to UI update)	p95 under 1 second	Performance trace
Candidate understands failed condition	85% in usability test (5 users, task-based)	Moderated testing
Pathway availability for common gaps	90% of gaps map to at least one catalogue course	Catalogue coverage report

05 Core Principle: Eligibility vs Preference

This is the heart of RozgarLink. Every attribute of an opportunity is classified at entry time as either an **Eligibility Criterion** or a **Preference Attribute**. The classification is explicit, visible and cannot be inferred by AI.

	Eligibility criterion (mandatory)	Preference attribute (ranking only)
Examples	Minimum qualification, residence state/district, age range, required certification, minimum experience	Commute distance, pay range, shift, work type, sector interest
Effect if not met	Opportunity is NOT an eligible match, regardless of skill fit	Opportunity stays eligible; rank is lower
Defined by	The opportunity (officer enters it)	Candidate states preferences; opportunity supplies attribute values
Uses AI?	Never	Inputs may be AI-informed (skill similarity)
Appears in UI as	Pass / Fail / Unknown chips with exact condition	'Why this ranks here' score breakdown

Three-valued evaluation

Result	Meaning	Where the opportunity appears
PASS	Candidate data satisfies the criterion	Counts toward eligibility
FAIL	Candidate data present and does not satisfy it	Opportunity moves to Not eligible, with the failing criterion named
UNKNOWN	Candidate data required is missing or unverified	Opportunity moves to Need more information

Opportunity status = ELIGIBLE if all mandatory criteria PASS; NOT_ELIGIBLE if any FAIL; NEEDS_INFO if none FAIL but at least one UNKNOWN.

Decision flow

```
for each opportunity O, for candidate C:
  results = [ evaluate(rule, C) for rule in O.eligibilityRules ]
  if any(r == FAIL):        status = NOT_ELIGIBLE   (list all failed rules)
  elif any(r == UNKNOWN):   status = NEEDS_INFO     (list missing fields)
  else:                     status = ELIGIBLE       -> goes to ranking
ranking(O, C) is called ONLY when status == ELIGIBLE
```

Hard architectural guarantee
The ranking module cannot be imported by, and has no write access to, the eligibility module. Eligibility takes only structured profile facts as input; the AI skill layer outputs are not in its type signature. A unit test fails the build if this dependency appears.

Edge cases

- A candidate with a perfect skill match but missing certification: NOT_ELIGIBLE, with message 'Requires Fire Safety Certificate. You have: none on profile.'
- A required skill is *not* an eligibility criterion unless the officer explicitly adds a 'Certification required' rule; skill shortfalls affect rank and appear as gaps.
- Multiple failures are all listed, ordered by candidate effort to fix (qualification, certification, experience, then residence, age).

06 Functional Requirements

ID	Requirement	Priority
FR-01	Candidate registers/logs in and creates a profile with skills, qualifications (level, field, year), location (state, district), and preferences	Must
FR-02	Candidate may enter skills as tags or free text; system extracts canonical skills and shows them for confirm/edit	Must
FR-03	Officer creates opportunities with title, description of work involved, location, required skills (with importance), optional skills	Must
FR-04	Officer defines eligibility criteria via a rule builder; each rule has type, operator, value and a plain-language label	Must
FR-05	Officer defines preference attributes (pay, shift, work type, sector) for ranking	Must
FR-06	System evaluates all eligibility rules deterministically; AI output is never an input	Must
FR-07	System ranks only eligible opportunities using skill fit, preference fit and proficiency	Must
FR-08	System recognises skills described in different words via alias dictionary then AI similarity	Must
FR-09	Every eligible result displays plain-language reasons for its rank	Must
FR-10	Every not-eligible result names each failed condition and the candidate's current value	Must
FR-11	Opportunities with missing candidate data appear under Need more information with the missing field named	Must
FR-12	Candidate marks interest; system shows missing required skills (gap analysis)	Must
FR-13	System suggests learning pathways from the supplied catalogue; sequencing respects prerequisites	Must
FR-14	Any change in qualification, location, skills or preferences recomputes and updates results immediately with visible change markers	Must
FR-15	Candidate can adjust preference weights (e.g. commute importance) and see the rank effect	Should
FR-16	Officer sees preview: number of sample candidates eligible for the opportunity as rules are edited	Should
FR-17	Catalogue manager maintains skills, aliases, courses, prerequisites via UI and CSV import	Must
FR-18	Audit log of eligibility decisions, rule changes and AI suggestions accepted/rejected	Must
FR-19	'What if' view: candidate previews eligibility if they gained a qualification or moved (not saved)	Could
FR-20	Exportable match report (PDF) for a candidate to take to a counsellor	Could

07 Data Model

Entity	Key fields	Notes
User	id, role, email/phone, name, language	Roles: candidate, officer, catalogue_manager, admin, evaluator
Candidate	id, userId, state, district, pincode, dob, preferences(json), updatedAt	One-to-one with user
Qualification	id, candidateId, level(ordinal), field, institution, year, verified(bool)	Level maps to configured ladder
Certification	id, candidateId, name, issuer, validTill	Matched by canonical certification code
CandidateSkill	candidateId, skillId, proficiency(1-5), source(manual|ai), confirmed(bool), confidence	AI-suggested skills need confirmation
Skill	id, canonicalName, category, description	Taxonomy
SkillAlias	id, skillId, phrase, language	Dictionary layer
Opportunity	id, title, workDescription, state, district, status, closesOn, createdBy	Entered manually
OpportunitySkill	opportunityId, skillId, importance(required|preferred), weight	Skill fit input
EligibilityRule	id, opportunityId, type, operator, value(json), label	Mandatory; deterministic
PreferenceAttribute	opportunityId, key(pay|shift|type|sector), value	Ranking only
LearningResource	id, title, provider, durationHours, level, mode, url(optional), cost	Team-supplied catalogue
ResourceSkill / Prerequisite	resourceId, skillId, depth / resourceId, requiresResourceId	Pathway graph
MatchResult (cache)	candidateId, opportunityId, status, score, explanation(json), computedAt	Rebuildable; never source of truth
AuditEvent	id, actor, type, entity, before, after, at	Immutable, append-only

Key relationships

A Candidate has many Qualifications, Certifications and CandidateSkills. An Opportunity has many EligibilityRules, OpportunitySkills and PreferenceAttributes. A LearningResource teaches many Skills and may require other LearningResources. MatchResult is derived and can be deleted and recomputed at any time.

Qualification ladder (configurable example)

Ordinal	1	2	3	4	5	6	7
Level	8th	10th	12th	ITI	Diploma	Graduate	Postgraduate

Rule 'minimum qualification = Diploma' passes for ordinal 5 or above. Field-specific rules (e.g. 'ITI in Electrician') use level plus a field code. Equivalences (e.g. ITI counts as 12th for a given opportunity) are explicit rules, never guessed.

08 Eligibility Engine Specification

Rule type	Operators	Value example	Candidate data used	Plain-language label
QUAL_MIN_LEVEL	gte	Diploma	Highest verified/stated qualification	Requires at least a Diploma
QUAL_FIELD	in	ITI: Electrician, Wireman	Qualification level and field	Requires ITI in Electrician or Wireman
RESIDENCE_STATE	in	Maharashtra	Candidate state	Open to residents of Maharashtra
RESIDENCE_DISTRICT	in	Pune, Nashik	Candidate district	Open to residents of Pune or Nashik
AGE_RANGE	between	18-35	Computed from date of birth, as on reference date	Age 18 to 35 as on 1 Jan 2027
CERT_REQUIRED	has	Fire Safety Certificate	Valid certifications	Requires a valid Fire Safety Certificate
EXPERIENCE_MIN	gte	12 months in electrical	Work experience entries	Requires 12 months of electrical experience
ATTRIBUTE	eq / in	Admin-defined field	Custom profile attribute	Programme-defined; see note

Evaluation contract

```
type RuleResult = { ruleId, outcome: 'PASS'|'FAIL'|'UNKNOWN',
                    label, candidateValue, requiredValue, missingField? }
evaluate(rule: EligibilityRule, profile: CandidateFacts): RuleResult   // pure, no I/O, no AI
evaluateOpportunity(rules, profile) -> { status, results[] }
```

Behaviour requirements

- **Pure and deterministic:** same inputs always give the same output; no randomness, no network calls, no model calls.
- **Versioned rules:** editing a rule creates a new rule version; MatchResult stores the version it used.
- **Reference date:** age and validity rules use an explicit date stored on the opportunity, so results are reproducible.
- **Unknown handling:** if the needed candidate field is empty, return UNKNOWN with the missing field.
- **Exhaustive reporting:** evaluate all rules (no short-circuit) so the candidate sees every failure at once.
- **Verification flag:** programme may require 'verified' qualifications; unverified ones return UNKNOWN for rules marked requiresVerification.

Sensitive attributes
Rules on sensitive personal attributes (for example reservation category) are disabled by default. They can be enabled only by a programme admin where the opportunity's governing scheme requires them, with the field marked sensitive, access-controlled and excluded from AI prompts and analytics.

09 Skill Normalisation and AI Guardrails

Three-layer skill recognition

Layer	Method	Output	Used for
1. Dictionary	Normalise text (lowercase, strip punctuation, stem), exact/fuzzy match against Skill and SkillAlias	Canonical skill, confidence 1.0	Ranking and display
2. Embedding match	Embed phrase; nearest canonical skills by cosine similarity above threshold	Candidate skill list + similarity	Ranking (partial credit)
3. LLM judgement	Prompt with the phrase and shortlist; return structured JSON: skill, relation (same|related|unrelated), confidence, short reason	Validated suggestion	Suggest to candidate/officer

Relation credit used by ranking

Relation	Condition	Credit
Same skill (alias or confirmed)	Dictionary hit or LLM 'same' and similarity at least 0.85	1.00
Closely related	Similarity 0.70 to 0.85 or LLM 'related' with confidence at least 0.7	min(0.85, similarity)
Unrelated	Below 0.70	0

Guardrails (non-negotiable)

- **No AI in eligibility.** AI outputs are not accepted by any function in the eligibility module.
- **Human confirmation:** AI-extracted skills on a profile are flagged 'suggested' until the candidate confirms; unconfirmed ones earn at most 50% credit.
- **Structured outputs only:** LLM responses are validated against a JSON schema; invalid output is discarded and logged.
- **Cache and cost control:** phrase to skill resolutions are cached; repeated phrases never call the model again.
- **Graceful degradation:** if AI is down, layer 1 still works; new unknown phrases are queued and shown as 'we will review this skill'.
- **Learning loop:** catalogue manager reviews queued phrases weekly and promotes good ones to SkillAlias, improving the dictionary.
- **Privacy:** only skill phrases are sent to the AI service; no names, contact details or sensitive attributes.
- **Transparency:** the UI marks AI-assisted matches with a small label and the reason sentence.

Example

Candidate wrote	Resolved skill	Layer	Credit
advanced excel, ms-excel	Spreadsheet (Excel)	Dictionary	1.00
tally prime, gst returns	Accounting software; Tax filing	Dictionary	1.00
fixing home power points	Electrical wiring (related)	LLM + embedding, confirmed	0.85
playing cricket	No match	Embedding below threshold	0

10 Ranking Engine

Ranking runs only on opportunities whose status is ELIGIBLE. Score is 0 to 100 and is fully decomposable into the parts below so it can be explained.

```
Score = 100 x ( 0.55 x SkillFit + 0.30 x PreferenceFit + 0.15 x ProficiencyFit )
```

Component	Definition	Range
SkillFit	Weighted average over opportunity skills of relation credit. Required skills weight 3, preferred skills weight 1. Missing skill = 0	0 to 1
PreferenceFit	Average of preference sub-scores using candidate's weights (default equal): commute, pay, shift, work type, sector	0 to 1
ProficiencyFit	For matched skills, min(1, candidate proficiency / required proficiency), weighted as above; default 1 if none specified	0 to 1

Preference sub-scores

Preference	Rule
Commute	1.0 within preferred km; linear fall to 0 at 2x preferred; 0.5 if remote/hybrid allowed and candidate open to it
Pay	1.0 if offered at or above expected minimum; proportional below; 0.5 if pay not stated
Shift / work type / sector	1.0 match, 0.5 flexible or unspecified, 0.0 mismatch (never excludes the opportunity)

Tie-breakers and rules

- Ties broken by: higher SkillFit, then nearer closing date, then newer posting.
- Weights live in configuration, versioned, and shown in the explanation ('skills 55%, preferences 30%, proficiency 15%').
- Candidates may re-weight preferences only; skill weight is fixed to protect fairness and explainability.
- A preference mismatch can never remove an eligible opportunity from the list; it can only lower its rank.
- Ranking is deterministic given the same inputs and AI relation credits (cached), so repeated views are stable.

Worked example

Opportunity (eligible)	SkillFit	PrefFit	ProfFit	Score	Reason in words
Electrician Apprentice, Pune	0.92	0.85	1.00	91	Matches all required skills; 8 km commute within your 15 km; pay meets target
Wiring Technician, Pimpri	0.78	0.60	0.90	74	One related skill (partial credit); night shift differs from preference

11 Explanation Engine

Explanations are generated by **templates over structured data** (rule results and score components), not free-form AI text, so they are always accurate and consistent. An optional language polish step may rephrase, but cannot add facts.

Eligible match explanation template

```
Ranked #{rank} (score {score}/100) because:
  + Skills: you have {n_exact} of {n_required} required skills ({list}); {related_note}
  + Preferences: {commute_sentence}; {pay_sentence}; {shift_sentence}
  - Gaps: {missing_required_skills or 'none'}
You meet all eligibility conditions: {criteria_list}
```

Not-eligible explanation template

```
Not eligible: {failed_label}.
Your profile: {candidate_value}.   Required: {required_value}.
How you could qualify: {fix_hint or 'not changeable'}
```

Examples

Case	Message shown
Eligible, rank 1	Ranked #1 (91/100): you have 4 of 4 required skills. 'Fixing home power points' was matched to Electrical wiring and confirmed by you. Commute is 8 km, within your 15 km preference. You meet all conditions: ITI Electrician, resident of Maharashtra, age 22.
Not eligible - residence	Not eligible: open to residents of Nashik and Pune districts only. Your profile: district Satara. This cannot be changed by learning; update your location if it is out of date.
Not eligible - qualification	Not eligible: requires at least a Diploma. Your profile: 12th pass. A Diploma in Electrical Engineering would meet this.
Not eligible - two failures	Two conditions not met: (1) requires Fire Safety Certificate, you have none; (2) age limit 18-30, you are 32.
Needs info	We cannot check this yet: requires a valid driving licence. Add your licence details to see if you are eligible.

Requirements

- Reasons must name the **specific** condition, never generic text like 'does not meet criteria'.
- Reading level target: Grade 8; no jargon such as 'embedding' or 'threshold' in candidate-facing text.
- All strings are keyed for translation (English, Hindi, Marathi) with the same placeholders.
- Explanations are stored with MatchResult and the rule/weight versions used, for audit.

12 Skill Gaps and Learning Pathways

Gap analysis (triggered when candidate marks 'Interested')

- Compare the opportunity's required and preferred skills to the candidate's confirmed skills (with relation credit below 0.85 counted as partial gap).
- Gap list shows each missing skill, importance, and whether it is a full gap or a partial gap ('you have related experience').
- Gap analysis also runs for Not eligible opportunities the candidate is interested in, but it shows eligibility blockers separately from skill gaps, because courses cannot fix residence.

Learning catalogue (supplied by the team)

Field	Description
id, title, provider (text label)	Provider is a name only; no live access assumed
skillsTaught[]	Canonical skill ids with level reached
prerequisites[]	Other catalogue items or skills required first
durationHours, mode, language, cost	Used for sorting and filters
url (optional)	Static link opened by the candidate; system never calls it
qualificationAwarded (optional)	Lets pathways suggest ways to meet eligibility, e.g. a Diploma or certificate

Pathway algorithm

```
missing = requiredSkills - candidateSkills                    (partial gaps included)
candidates = catalogue items teaching any skill in missing
choose a minimum-cost cover of 'missing' (greedy set cover, tie-break: fewer hours, lower cost)
expand prerequisites by topological order, skipping ones candidate already has
output ordered steps with total hours and 'skills you will gain' per step
```

Pathway output

Step	Course	Hours	Skills gained	Why
1	Basics of Industrial Safety	10	Safety procedures	Prerequisite for step 2
2	Panel Wiring and Fault Finding	40	Panel wiring, Fault finding	Closes 2 of 2 missing required skills

- If no catalogue item teaches a missing skill, the UI says 'No course in our catalogue yet' and logs it for the catalogue manager.
- Eligibility blockers with a qualification route (e.g. Diploma) are shown as 'To become eligible' pathways using qualificationAwarded.
- Pathways recompute when the candidate marks a course as completed, which adds the skill as 'learned (self-reported)' pending verification.

13 Real-Time Updates

Requirement: when a candidate changes qualifications or location (and also skills or preferences), eligibility and rankings update **immediately and visibly**.

Event flow

Step	Action	Target time
1	Candidate saves a change (e.g. adds ITI qualification). API validates and writes in one transaction	under 100 ms
2	Change triggers recompute job for that candidate only (not the whole system)	under 50 ms
3	Engine prefilters opportunities (open, relevant geography) then evaluates rules in memory	under 300 ms for 10k opportunities
4	Ranking and explanations built for eligible ones; MatchResult upserted	under 300 ms
5	Server-Sent Event 'matches.updated' with diff pushed to open sessions	under 100 ms
6	UI animates changes and shows a change banner	immediate

Visible change markers

Change	UI treatment
Became eligible	Card moves up with 'New: now eligible' badge and the rule that flipped (e.g. 'You now meet: ITI Electrician')
No longer eligible	Card moves to Not eligible with 'Changed' badge and the failed rule
Rank moved	Arrow with delta, e.g. '+3 places'; explanation refreshed
Needs info resolved	Card leaves 'Need more information' into the correct list
Summary banner	'Your change updated 6 results: 2 newly eligible, 1 no longer eligible, 3 re-ranked' with Undo (restores previous profile value)

Consistency requirements

- The save call returns only after recompute finishes for the optimistic result, so a refresh never shows stale state.
- Opportunity or rule edits by an officer trigger recompute for affected candidates in a background queue; candidates with an open session receive a push.
- Rapid successive edits are debounced (150 ms) and collapsed; the latest state always wins.
- If push fails, the client polls /matches?since=version every 5 seconds as a fallback.
- Every result carries profileVersion so the UI can detect and discard out-of-order updates.

14 UI/UX Specification (FutoraOne Design Language)

RozgarLink ships inside the **FutoraOne** app shell: mobile-first, white rounded cards on a soft grey-blue canvas, pill filters, blue primary actions, a four-tab bottom bar (**Home, Match, Learn, Profile**) and a floating **AI** button. Same patterns, new content.

FutoraOne screen	RozgarLink content	Key elements
Home	'Your Career Path Starts Here' hero, search (opportunities, skills, courses), quick tiles: Match, Learn, Explore, Profile	Recommended for you = top eligible matches; Top Matches strip; profile-strength nudge
Match	Header 'Match' with pill filters: **All, Eligible, Need info, Not eligible**	Card = title, location, 'Eligible' green dot, score, top 3 reasons, skill chips, **Interested** button (replaces Connect); Not-eligible cards show the failed rule in a red-tinted strip, no score
Opportunity detail	Work involved; two blocks: 'You must have' (eligibility) and 'These affect ranking' (preferences); your status per rule	Reinforces the eligibility vs preference split
Learn	Pill tabs **Courses, Paths, Practice**; course-path cards with progress bar and 'x/y modules'	Paths = gap pathways from the catalogue; 'Mark completed' adds a self-reported skill
Profile	Stats row: Courses completed, Skills confirmed, Interested; profile-strength ring; learning streak	Menu: My Courses, Saved, Settings, Help & Support; inline edit of qualification/location with instant recompute and Undo
AI button (floating)	Ask AI: 'Why am I not eligible?', 'What should I learn?', skill extraction from free text	Explains stored rule results; **never** changes eligibility; AI-assisted items carry a small label
Officer console (web)	Opportunity form, rule builder, eligible-candidate preview, catalogue manager, audit, admin	Desktop layout using the same cards, pills and blue actions

UX requirements

- Colour is never the only signal: Pass/Fail/Unknown use icons and words as well as colour.
- Failed conditions appear as a plain sentence at the top of an ineligible card, not hidden in details.
- Eligible and Not eligible lists are visually separate; scores never appear on ineligible cards.
- Accessibility WCAG 2.1 AA, 44 px tap targets, screen-reader labels, light on bandwidth (initial JS under 300 KB).
- English plus Hindi and Marathi string files ready; empty states guide the next action ('Add your qualification to unlock 4 more opportunities').
- Live change banner after any profile edit: '2 newly eligible, 1 no longer eligible, 3 re-ranked' with Undo.

Key user flow

```
Sign in -> Complete profile -> Match loads (eligible ranked, not eligible with reasons)
-> Tap card -> why / failed rule -> Interested -> gap + Learn path
-> Edit location or qualification -> results change instantly with banner and badges
```

15 System Architecture and Technology

```
[ Web app: Next.js + TypeScript + Tailwind ]
          |  REST + Server-Sent Events
          v
[ API: Node.js + Express + TypeScript ]
   |-- Auth and RBAC middleware
   |-- Profile service      |-- Opportunity service
   |-- Matching service:  eligibility-engine (pure) | ranking-engine | explainer
   |-- Skill service: dictionary + embeddings + LLM client + cache
   |-- Pathway service (catalogue graph)
   |-- Event bus (in-process / Redis pub-sub) -> SSE gateway
          |
          v
[ PostgreSQL + pgvector ]   [ Redis cache / queue ]   [ External LLM / embedding API ]
```

Layer	Choice	Reason
Frontend	Next.js, React, TypeScript, Tailwind	Company standard stack; SSR for fast low-bandwidth loads
Backend	Node.js, Express, TypeScript, Zod validation	Shared types with frontend; schema-validated inputs
Database	PostgreSQL with pgvector (or in-memory vectors for MVP)	Relational integrity for rules; vector search for skills
Cache and queue	Redis (BullMQ)	Skill phrase cache, background recompute
Realtime	Server-Sent Events (WebSocket optional)	Simple, one-way updates, proxy friendly
AI	Claude API for extraction and relation judgement, embeddings model for similarity	Structured JSON output, swap-able provider interface
Hosting	Containerised (Docker), cloud region in India	Data residency; easy scaling
Observability	Structured logs, OpenTelemetry traces, Sentry	Trace any decision

Module boundaries

- **eligibility-engine**: pure TypeScript package, zero dependencies on AI, DB or network; 100% unit-test coverage target.
- **ranking-engine**: receives only ELIGIBLE opportunities and cached relation credits.
- **explainer**: templates plus i18n catalogue; deterministic.
- **skill-service**: only component that talks to the AI provider; returns structured, validated results.
- **match-orchestrator**: coordinates the above, writes MatchResult, publishes events.

Scaling notes

Prefilter by state/district, closing date and status using indexed columns before in-memory rule evaluation. For 10k opportunities, evaluation is a tight loop of simple comparisons. Recompute is per candidate; officer rule changes enqueue affected candidates in batches.

16 API Specification (REST + SSE)

Method and path	Purpose	Notes
POST /auth/login, /auth/otp	Authenticate	JWT with role claim
GET/PUT /candidates/me	Read/update profile	Triggers recompute; returns profileVersion
POST /candidates/me/qualifications	Add qualification	Also PATCH/DELETE by id
POST /candidates/me/skills/extract	Free text to suggested canonical skills	AI-assisted; returns confidence and reason
PUT /candidates/me/skills	Confirm/edit skills	Sets confirmed=true
GET /matches	Eligible, needsInfo, notEligible with explanations	Query: since, filters, weights
GET /matches/{opportunityId}	Full per-rule breakdown	Includes rule results and score parts
POST /matches/whatif	Preview with hypothetical profile changes	Not persisted
POST /interests	Mark Interested	Returns gap analysis
GET /opportunities/{id}/gap	Skill gap and pathway	Uses catalogue
POST/PUT /opportunities	Create/update opportunity with rules	Officer only; validates rules
POST /opportunities/{id}/preview-eligibility	Sample-candidate eligibility count	Officer tool
CRUD /skills, /aliases, /courses	Catalogue management	Catalogue manager; CSV import endpoint
GET /audit	Search audit events	Admin, evaluator
GET /events (SSE)	Stream matches.updated, catalogue.updated	Auth by token

Sample response: GET /matches

```
{ "profileVersion": 42,
  "eligible": [ { "opportunityId": "OP-117", "rank": 1, "score": 91,
       "reasons": ["Has 4 of 4 required skills", "Commute 8 km within preference"],
       "criteria": [ {"label":"ITI Electrician","outcome":"PASS"} ],
       "gaps": [] } ],
  "notEligible": [ { "opportunityId": "OP-203",
       "failed": [ {"label":"Open to residents of Nashik or Pune","candidateValue":"Satara",
                    "outcome":"FAIL"} ] } ],
  "needsInfo": [ { "opportunityId": "OP-301", "missingField": "drivingLicence" } ],
  "changes": { "newlyEligible": ["OP-117"], "noLongerEligible": [], "rankMoved": 3 } }
```

- All inputs validated with Zod; all endpoints rate limited; errors use RFC 7807 problem details.
- Idempotency key supported on POST requests that create data; API versioned under /v1.

17 Non-Functional Requirements

Area	Requirement
Performance	Recompute p95 under 1 s end-to-end; matches list loads under 2 s on 3G; supports 500 concurrent sessions at MVP
Availability	99.5% monthly; AI outage must not stop matching (degrades to dictionary-only)
Security	OWASP ASVS L2; JWT with short expiry and refresh; RBAC on every endpoint; rate limiting; secrets in vault; dependency scanning
Privacy and compliance	Align with India's Digital Personal Data Protection Act 2023: purpose limitation, consent capture, data export and delete on request, minimal data to AI vendor, India data residency
Auditability	Append-only AuditEvent for rule edits, eligibility decisions, AI suggestions and overrides; retained 3 years; searchable by candidate and opportunity
Fairness	No use of gender, religion, caste or disability in ranking; sensitive eligibility fields only where scheme demands; quarterly review of outcome distribution by district and qualification level
Explainability	Every result reproducible from stored inputs, rule versions, weights version and cached AI relation credits
Accessibility	WCAG 2.1 AA; screen-reader and keyboard friendly; plain-language content
Localisation	String catalogue for English, Hindi, Marathi; right-to-left not required at MVP
Maintainability	TypeScript strict mode; CI with lint, type-check, unit, integration and E2E tests; infra as code
Data quality	Opportunity form blocks conflicting rules (e.g. age 30-20), flags missing eligibility; skill dictionary reviewed weekly
Backup and recovery	Daily backups, RPO 24 h, RTO 4 h; MatchResult can be rebuilt from source of truth

Human oversight

RozgarLink is decision support. Screens display 'Eligibility follows the criteria stated by the opportunity; confirm with the issuing office before applying.' Officers can correct an opportunity's rules at any time; candidates can contest a result by flagging it, which creates a review ticket with the full decision trace attached.

18 Test Plan and Acceptance Criteria

Acceptance criteria mapped to the challenge

Challenge requirement	Acceptance test	Pass condition
Eligibility separate from preference ranking	AT-01: candidate with 100% skill match fails residence rule	Opportunity in Not eligible; never in ranked list
Rank only eligible opportunities	AT-02: inspect ranking input in 1,000 randomised profiles	Zero non-eligible items scored
Different wording, same skill	AT-03: 300 labelled phrases	At least 95% resolve correctly; failures logged
AI must not decide eligibility	AT-04: disable AI entirely; compare eligibility outputs	Identical eligibility results with and without AI
Explain eligible rank	AT-05: all eligible results include reasons	100% coverage, readable, correct figures
Explain failed condition	AT-06: each failed rule shown with candidate value	Specific rule label present for 100%
Skill gaps and pathways	AT-07: mark interested; compare to catalogue	Gaps correct; pathway respects prerequisites; no network calls
Immediate visible updates	AT-08: change qualification and location	UI updates under 1 s with banner and badges
Decision support	AT-09: disclaimer and audit trail present	Decision trace retrievable for any result

Test layers

Layer	Scope	Tooling
Unit	Every rule type with pass/fail/unknown boundaries (age on exact birthday, ladder boundaries); score components; explanation templates	Vitest, property-based tests (fast-check)
Contract	API schemas, i18n placeholders, SSE event shapes	Zod schemas, Pact-style tests
Integration	DB, queues, recompute, caching, AI client with mocked responses	Testcontainers
AI evaluation	Alias and relation accuracy on labelled set; prompt regression	Offline eval harness each release
E2E	Candidate and officer journeys, including live update scenarios	Playwright
Non-functional	Load, accessibility, security scans	k6, axe, ZAP

Critical guard tests (block release)

- Eligibility module has no import path to AI or ranking modules (static dependency check).
- Property test: for any profile, any opportunity with a FAIL rule is never returned in eligible list.
- Determinism test: same inputs give byte-identical eligibility output across 100 runs.

19 Seed Data and Demo Scenarios

Seed dataset (for demo and tests)

Dataset	Size	Contents
Skills	40 canonical	Electrical wiring, Panel fitting, Spreadsheet, Accounting software, Tax filing, Tailoring, CNC basics, Customer service, Data entry, etc.
Aliases	200+	'ms excel', 'advanced excel', 'tally', 'gst filing', 'stitching', 'sewing machine operation', 'mcb installation' ...
Opportunities	30	Apprenticeships and jobs across Maharashtra districts with mixed rules and preferences
Candidates	6	Asha, Rahul, Meena and three edge-case profiles (missing data, borderline age, unverified qualification)
Courses	20	Safety basics, Panel wiring, Excel for work, Tally with GST, Diploma bridging, Spoken English, etc., with prerequisites

Demo scenarios (must work end to end)

#	Scenario	Expected outcome
A	Asha moves her district from Pune to Satara	Two residence-restricted roles flip to Not eligible instantly; banner reports changes; explanation names 'Open to residents of Pune or Nashik'
B	Rahul adds a Diploma	Two Diploma-required roles become eligible and appear ranked with 'New: now eligible' badges
C	Alias test: Rahul writes 'tally prime, gst returns'	Mapped to Accounting software and Tax filing; matched without keyword overlap; marked as dictionary match
D	Meena sets commute importance to high	Order of eligible list changes; nothing disappears; explanation updates
E	Candidate with perfect skills but too old	Role is Not eligible with 'age limit 18-30, you are 32'; rank not computed
F	Mark 'Interested' on role needing two missing skills	Gap list plus 2-step pathway with prerequisites and total hours
G	Missing licence data	Role appears in Need more information with the missing field
H	AI service switched off	Dictionary matching and all eligibility work unchanged; new phrase queued for review
I	Officer changes a rule on a live opportunity	Affected candidates receive push; lists update; audit log shows before/after

Demo script (8 minutes)
1) Show opportunity form with rule builder 2) Log in as Asha, view matches 3) Change district, watch live update 4) Open a not-eligible card 5) Mark interested, show pathway 6) Add Diploma as Rahul 7) Turn AI off, show eligibility unchanged 8) Show audit trace.

20 Roadmap, Risks and Open Questions

Delivery plan (10 weeks)

Phase	Weeks	Deliverables	Exit criteria
0. Foundations	1	Repo, CI, schema, auth/RBAC, design system	Hello-world deploy
1. Core engine	2-4	Eligibility engine, rule builder, ranking, explainer, seed data	AT-01, AT-02, AT-04, AT-05, AT-06 pass
2. Skills and AI	4-6	Dictionary, embeddings, LLM extraction, confirmation UI, cache	AT-03 at 95% or higher
3. Gaps and pathways	6-7	Catalogue management, gap analysis, pathway builder	AT-07 pass
4. Live updates	7-8	Recompute pipeline, SSE, change markers, what-if	AT-08 pass; p95 under 1 s
5. Hardening	9-10	Accessibility, security, load test, audit, i18n, demo polish	All acceptance tests green; pilot ready

Risks and mitigations

Risk	Impact	Mitigation
AI maps unrelated skills as the same	Poor ranking	Confidence thresholds, candidate confirmation, weekly review, capped credit
Officers enter rules wrongly	Wrong eligibility	Rule builder validation, plain-language preview, second-officer approval option, audit
Candidate gives inaccurate data	Wrong results	Verification flag, disclaimers, 'confirm with issuing office' notice
Catalogue lacks courses	Weak pathways	Gap queue to catalogue manager; show honest 'no course yet'
Ambiguous equivalence (e.g. ITI vs 12th)	Wrong pass/fail	Explicit equivalence rules per opportunity, never inferred
Bias in ranking	Unfair outcomes	No protected attributes in ranking; quarterly distribution review
AI vendor outage or cost spike	Reduced skill matching	Dictionary fallback, cache, rate budget

Open questions for stakeholders

- Which qualification ladder and equivalences does the programme officially recognise?
- Are any sensitive-attribute eligibility rules legally required for the first schemes, and who approves them?
- Is candidate qualification self-declared in MVP, or must it be verified against documents?
- Which languages are mandatory at launch, and who supplies translations?
- What is the format of the team-supplied learning catalogue (CSV, sheet), and who owns updates?

Definition of done
All 20 requirements of the challenge demonstrated through acceptance tests AT-01 to AT-09, scenarios A to I run live, audit trace retrievable for any result, and zero ineligible opportunities ever shown as eligible.